

Weekly Project Progress Report & Deployment Preparation

1. Project Overview

Project Title: TaskFlow — Real-Time Collaborative Task Management Platform with Sentiment-Aware Feedback Analysis

Short Description: TaskFlow is a full-stack MERN web application that enables users to manage personal and project-based tasks, collaborate with teammates in real time, track audit logs, and gain intelligent insights from user feedback through a built-in NLP sentiment analysis engine.

Current Completion: 100% — Fully deployed and live

Live URL: <https://github-publish.replit.app>

Demo Video: <https://drive.google.com/file/d/1gp-gPIHlvm6GAQ9Tfdmhyx5xqWWj0YOj/view?usp=drivesdk>

2. Team Contributions

Member 1: Mostafa Tfaily (120080)

Tasks Completed:

- Designed and implemented all Mongoose schemas (User, Task, Project, Friendship, Notification, Log, Feedback)
- Built all backend routes: users, tasks, projects, friendship, notifications, logs, feedbacks
- Implemented JWT authentication middleware and session security
- Engineered real-time Socket.io server — room management, event emission for friend requests, project invitations, task assignments, and accepted requests
- Implemented audit logging system for full traceability of all user and admin actions
- Integrated Nodemailer for password reset and change-password confirmation emails
- Implemented all API data fetching and integration on the frontend — connecting every page to its backend endpoint with proper error handling, loading states, and auth headers
- Implemented the Sentiment Analysis engine from scratch in TypeScript (VADER-inspired weighted lexicon, negation detection, score normalization — no external library)

Contribution:

Full backend architecture, Socket.io real-time layer, audit logs, and all frontend API integration logic.

Member 2: Wassem Abou Arab (120581)

Tasks Completed:

- Designed and implemented the full UI of all application pages using React, Tailwind CSS, and custom dark-theme styles
- Built reusable frontend components: Navbar, Sidebar, Footer, TaskStructure, ProtectedRoute
- Designed the friends interface, project dashboard UI, admin dashboard layout, and all form pages
- Prepared the frontend project environment and component structure for backend integration

Contribution:

Frontend UI design and implementation — all visual layouts, component architecture, and user experience.

3. Work Completed (Full Feature List)

- User registration, login, and JWT-based authentication
- Forgot / reset password flow via email (Nodemailer)
- Change password from Profile with bcrypt verification and email confirmation
- Personal task CRUD — create, edit, delete with priority, due date, start/end time
- Done / Undone / All task views with priority filtering
- Project management with owner and member roles
- Assign tasks to project members with priority, description, and due date
- Friends system — send, accept, and block friend requests
- Real-time notifications via Socket.io (friend requests, project invitations, task assignments, accepted requests)
 - Global Socket context wrapping the entire app for live event propagation
 - Analytics dashboard with circular progress charts (react-circular-progressbar)
 - Admin dashboard — manage users, update roles, ban/unban, delete accounts
 - Admin username filter input for fast user lookup
 - Registration Date column in admin user table
 - Audit logs page — full traceability for both user and admin actions
 - User feedback submission and admin review
 - Sentiment Analysis on user feedback — VADER-inspired NLP algorithm with weighted lexicon (~60 words, " 3 to +3), negation detection, score normalization, live badges, and summary panel
 - Profile management — edit username and email
 - Full deployment on Replit with MongoDB Atlas, environment secrets, and Socket.io proxying

4. Demonstration

Demo Video: <https://drive.google.com/file/d/1gp-gPIHlvm6GAQ9Tfdmhyx5xqWWj0YOj/view?usp=drivesdk>

5. Challenges & Solutions

Challenge	Resolution
Socket.io rooms not receiving events cross-route	Used a global SocketContext with useState so the socket instance persists across all pages via React context
White dropdown popup in native <select> elements	Applied solid hex #0b1a12 background with color-scheme: dark via global CSS — rgba falls back to white in native dropdowns
Aggregation for audit logs and analytics	Resolved using Mongoose populate + manual aggregation on backend routes
Bi-directional friendship prevention	Enforced in the addfriend route by checking both directions of the relation before inserting
MongoDB Atlas connectivity from Replit	Set Network Access to 0.0.0.0/0 in Atlas to allow all IPs

6. Deployment Checklist

- ' Application runs without errors
- ' MongoDB Atlas connected and seeded
- ' All environment variables configured

(MONGODB_URI, JWT_KEY, SESSION_SECRET,
MAIL_USER, MAIL_PASS)

- Backend deployed and serving at /api
- Frontend build deployed and serving at /
- Socket.io proxied correctly through /api/socket.io
- Full end-to-end testing completed
- Live at <https://taskflow.replit.app>